GEOMETRIC PHASE-SPACE COMPLETION AND CONDITIONAL GLOBAL REGULARITY FOR THE 3D INCOMPRESSIBLE NAVIER–STOKES EQUATIONS

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Abstract. Let $\mathbf{u}_0 \in C^\infty(\mathbb{T}^3;\mathbb{R}^3)$ be a smooth, divergence-free initial velocity field on the flat three-dimensional torus $\mathbb{T}^3 = \mathbb{R}^3/(2\pi\mathbb{Z})^3$. We formulate a geometric completion of Arnold's Lie–Poisson framework on $\mathrm{SDiff}(\mathbb{T}^3)$ via a three-legged metriplectic architecture: (1) the skew-symmetric Arnold Lie–Poisson bracket governing ideal transport, (2) a symmetric metriplectic dissipation sector incorporating viscous damping alongside a degenerate projected metric correction $B(F,G)$ (the Frohmanian Symplectic Tether), and (3) Marsden–Weinstein–Ratiu coadjoint orbit reduction. Because $B(F,H) \equiv 0$ for the kinetic-energy Hamiltonian H, the generated dynamics recover the exact, unmodified 3D Navier–Stokes vorticity transport equation. Within this framework we record three rigorous regularity results: a sharp conditional global continuation theorem via critical $L^1_t L^\infty_x$ full-gradient and strain integrability (Theorem A: Forms B-grad and B-strain), unconditional local-in-time smoothness on the deterministic Kato lifespan (Theorem B), and unconditional global smoothness for small critical $\dot{H}^{1/2}$ data (Theorem C). Finally, we isolate the supercritical $L^2_t L^2_x \to L^1_t L^\infty_x$ scaling gap as the precise open frontier for geometric vorticity-coherence analysis. No unconditional large-data claim is made; Clay Statement (B) remains open.

1. Statement of the Problem: The Clay Millennium Problem and the Supercriticality Barrier

1.1. **Classical PDE Posing (Fefferman Formulation).** Let $\mathbb{T}^3 = \mathbb{R}^3/(2\pi\mathbb{Z})^3$ denote the flat three-dimensional torus. The 3D incompressible Navier–Stokes equations in velocity-pressure form are

$$\begin{cases} \partial_t \mathbf{u} + (\mathbf{u}\cdot\nabla)\mathbf{u} + \nabla p = \nu\Delta\mathbf{u}, \\ \operatorname{div}\mathbf{u} = 0, \\ \mathbf{u}(x,0) = \mathbf{u}_0(x), \end{cases} \tag{1.1}$$

where $\mathbf{u}_0 \in C^\infty(\mathbb{T}^3;\mathbb{R}^3)$ is smooth and divergence-free, p is the scalar pressure, and $\nu > 0$ is the kinematic viscosity. Taking the curl of (1.1) eliminates the pressure gradient and yields the equivalent vorticity transport equation for $\boldsymbol{\omega} = \operatorname{curl}\mathbf{u}$:

$$\partial_t\boldsymbol{\omega} + (\mathbf{u}\cdot\nabla)\boldsymbol{\omega} = (\boldsymbol{\omega}\cdot\nabla)\mathbf{u} + \nu\Delta\boldsymbol{\omega}, \qquad \operatorname{div}\boldsymbol{\omega} = 0. \tag{1.2}$$

The Clay Millennium Prize Problem (Statement B) asks whether, for every smooth divergence-free $\mathbf{u}_0$, there exists a unique global classical solution $\mathbf{u} \in C^\infty(\mathbb{T}^3\times[0,\infty);\mathbb{R}^3)$ ($T^* = \infty$), or whether a finite-time singularity ($T^* < \infty$) can form [5].

1.2. **The Analytical Supercriticality Obstruction.** Testing (1.1) against $\mathbf{u}$ yields the unconditional global energy inequality, providing an $L^\infty_t L^2_x$ bound on velocity and an $L^2_t L^2_x$ bound on vorticity:

$$\|\mathbf{u}(t)\|^2_{L^2(\mathbb{T}^3)} + 2\nu\int_0^t \|\boldsymbol{\omega}(\tau)\|^2_{L^2(\mathbb{T}^3)}\,d\tau \le \|\mathbf{u}_0\|^2_{L^2(\mathbb{T}^3)}. \tag{1.3}$$

Date: November 9, 2026.

Under the natural parabolic rescaling $\mathbf{u}_\lambda(x,t) = \lambda \mathbf{u}(\lambda x, \lambda^2 t)$ on $\mathbb{R}^3$, the kinetic energy scales as

$$\|\mathbf{u}_\lambda(t)\|^2_{L^2(\mathbb{R}^3)} = \lambda^{-1} \|\mathbf{u}(\lambda^2 t)\|^2_{L^2(\mathbb{R}^3)} \to 0 \quad \text{as } \lambda \to \infty.$$

Because the scaling exponent is negative ($-1 < 0$), the L^2 energy bound is *supercritical*. Preventing finite-time blow-up requires controlling a scale-invariant (λ^0) critical norm, such as the $L^1_t L^\infty_x$ integrability of the symmetric strain tensor $S = \frac{1}{2}(\nabla \mathbf{u} + (\nabla \mathbf{u})^T)$, leaving a fundamental analytical gap between available $L^2_t L^2_x$ bounds and required $L^1_t L^\infty_x$ control.

2. Geometric Completion of Phase Space

Rather than modifying the underlying PDE, we embed (1.2) into a geometrically complete infinite-dimensional phase space over the group of volume-preserving diffeomorphisms $\mathrm{SDiff}(\mathbb{T}^3)$ with Lie algebra $\mathfrak{g} = \mathfrak{sdiff}(\mathbb{T}^3)$.

2.1. Leg 1: Reversible Lie–Poisson Sector.

On the dual space $\mathfrak{g}^*$, ideal inviscid fluid motion is governed by Arnold's skew-symmetric Lie–Poisson bracket [1], written here in the paper's vorticity identification

$$\{F, G\}_{\mathrm{LP}} = \int_{\mathbb{T}^3} \boldsymbol{\omega} \cdot \Big(\frac{\delta F}{\delta \boldsymbol{\omega}} \times \frac{\delta G}{\delta \boldsymbol{\omega}} \Big)\, dV. \tag{2.1}$$

Remark 2.1 (Jacobi identity by coadjoint reduction)**.** We do not derive the Jacobi identity for (2.1) from scratch. The bracket is the canonical reduced Poisson structure on $\mathfrak{sdiff}(\mathbb{T}^3)^*$, obtained by infinite-dimensional Marsden–Weinstein reduction of the geodesic (kinetic-energy) Hamiltonian flow on $T^*\mathrm{SDiff}(\mathbb{T}^3)$. Consequently the Jacobi identity holds as a theorem of Lie–Poisson / coadjoint-orbit reduction [1, 11, 2]. (An equivalent curl-form presentation $\int \boldsymbol{\omega} \cdot \big((\nabla \times \delta F/\delta \boldsymbol{\omega}) \times (\nabla \times \delta G/\delta \boldsymbol{\omega})\big)\, dV$ appears under the usual identification of variational derivatives with divergence-free vector fields; we retain (2.1) as the paper's written convention.)

Equipped with the kinetic-energy Hamiltonian

$$H[\boldsymbol{\omega}] = \frac{1}{2} \int_{\mathbb{T}^3} |\mathbf{u}|^2\, dV, \qquad \mathbf{u} = K_{\mathrm{BS}} * \boldsymbol{\omega},$$

where K_{BS} denotes the Biot–Savart kernel on $\mathbb{T}^3$, one has $\nabla \times (\delta H/\delta \boldsymbol{\omega}) = \mathbf{u}$ (equivalently $\delta H/\delta \boldsymbol{\omega} = \boldsymbol{\psi}$ with $\operatorname{curl} \boldsymbol{\psi} = \mathbf{u}$). The bracket $\{F, H\}_{\mathrm{LP}}$ generates the exact inviscid Euler dynamics tangent to coadjoint orbits $\mathcal{O} \subset \mathfrak{g}^*$.

2.2. Leg 2: Symmetric Metriplectic Sector and the Tether Correction.

To account for viscous dissipation ($\nu > 0$) while preserving exact algebraic symmetry, we equip $\mathfrak{g}^*$ with a symmetric, negative-semidefinite metriplectic bracket

$$[F, G]_{\mathrm{total}} = [F, G]_{\mathrm{diss}} + B(F, G),$$

defined by [12]

$$[F, G]_{\mathrm{diss}} = -\nu \int_{\mathbb{T}^3} \Big(\nabla \times \frac{\delta F}{\delta \boldsymbol{\omega}} \Big) \cdot \Big(\nabla \times \frac{\delta G}{\delta \boldsymbol{\omega}} \Big)\, dV, \tag{2.2}$$

$$B(F, G) = -\kappa \int_{\mathbb{T}^3} |\boldsymbol{\omega}|^2 \Big(\Pi_{\mathbf{u}} \Big(\nabla \times \frac{\delta F}{\delta \boldsymbol{\omega}} \Big) \cdot \Pi_{\mathbf{u}} \Big(\nabla \times \frac{\delta G}{\delta \boldsymbol{\omega}} \Big) \Big)\, dV, \tag{2.3}$$

where $\kappa = C_{\mathrm{CZ}}(3) > 0$ is the 3D Calderón–Zygmund constant and $\Pi_{\mathbf{u}}$ is the L^2-orthogonal projection onto $\mathrm{span}\{\mathbf{u}\}^\perp$:

$$\Pi_{\mathbf{u}} v = v - \frac{\langle v, \mathbf{u} \rangle_{L^2}}{\|\mathbf{u}\|^2_{L^2}} \mathbf{u} \quad (\mathbf{u} \neq 0).$$

Separating $\{F,G\}_{\mathrm{LP}}$ (skew-symmetric) from $[F,G]_{\mathrm{total}}$ (symmetric) leaves the Jacobi identity of the Poisson sector intact (Remark 2.1) while ensuring that $B(F,G)$ satisfies strict degeneracy with respect to the kinetic-energy Hamiltonian H:

$$\nabla\times\Big(\frac{\delta H}{\delta\boldsymbol{\omega}}\Big)=\mathbf{u}\implies\Pi_{\mathbf{u}}\Big(\nabla\times\frac{\delta H}{\delta\boldsymbol{\omega}}\Big)=\Pi_{\mathbf{u}}(\mathbf{u})\equiv 0\implies B(F,H)\equiv 0\quad\forall F. \tag{2.4}$$

Consequently, the complete metriplectic evolution

$$\frac{dF}{dt}=\{F,H\}_{\mathrm{LP}}+[F,H]_{\mathrm{total}}=\{F,H\}_{\mathrm{LP}}+[F,H]_{\mathrm{diss}} \tag{2.5}$$

reproduces the exact, unmodified 3D Navier–Stokes vorticity transport equation (1.2) without introducing artificial forcing terms. The correction B is *not* claimed to be Poisson.

2.3. **Leg 3: Marsden–Weinstein–Ratiu Reduction.** Because the volume form dV and the quadratic weight $|\boldsymbol{\omega}|^2\,dV$ transform covariantly under the coadjoint action of $\mathrm{SDiff}(\mathbb{T}^3)$, the reversible bracket (2.1) descends to a canonical symplectic structure on each coadjoint orbit $\mathcal{O}$ via Marsden–Weinstein–Ratiu reduction [11, 10, 2], while $[F,H]_{\mathrm{diss}}$ precisely parameterizes the transverse dissipative drift across orbits driven by kinematic viscosity.

3. Rigorous Regularity Theorems within the Framework

Within the metriplectic formulation (2.5), the regularity theory of 3D Navier–Stokes resolves into three complete theorems.

Theorem 3.1 (Conditional Global Regularity — Forms B-grad and B-strain)**.** *Let $\mathbf{u}_0\in H^s(\mathbb{T}^3;\mathbb{R}^3)$ ($s>5/2$) satisfy $\operatorname{div}\mathbf{u}_0=0$, and let $\mathbf{u}\in C([0,T^*);H^s(\mathbb{T}^3))$ be the maximal local Kato solution generated by (2.5). We record two formulations of the conditional hypothesis (Form B). Both are one-directional: they imply $T^*=+\infty$, but neither is claimed to follow from the energy inequality alone. Large-data Clay Statement (B) remains open.*
(B-grad) Full-gradient form (direct constant). *Assume that for every finite $T>0$ there exists a derived constant $C_*(\|\mathbf{u}_0\|_{H^s},\nu,T)<\infty$ such that*

$$\int_0^T\|\nabla\mathbf{u}(t)\|_{L^\infty(\mathbb{T}^3)}\,dt\le C_*(\|\mathbf{u}_0\|_{H^s},\nu,T). \tag{3.1}$$

Then $T^*=+\infty$, and the Kato–Ponce energy estimate [8]

$$\frac{d}{dt}\|\mathbf{u}(t)\|_{H^s}\le C_s\|\nabla\mathbf{u}(t)\|_{L^\infty}\|\mathbf{u}(t)\|_{H^s}$$

together with Gronwall's lemma yields the direct *bound*

$$\sup_{t\in[0,T]}\|\mathbf{u}(t)\|_{H^s}\le\|\mathbf{u}_0\|_{H^s}\exp\big(C_s\,C_*(\|\mathbf{u}_0\|_{H^s},\nu,T)\big). \tag{3.2}$$

(The same conclusion holds with the integrated form of Kato–Ponce as a cited input; see the companion Lean target `formB_Hs_bound`.) Smoothness for $t>0$ follows by parabolic bootstrap. The Beale–Kato–Majda / Kato continuation implication in $\nabla\mathbf{u}$

$$\int_0^T\|\nabla\mathbf{u}(t)\|_{L^\infty}\,dt<\infty\implies T^*>T$$

is cited [3, 8]*, not re-proved here.*

(B-strain) Strain-only form (log-Sobolev loss). *Assume instead that for every finite $T > 0$ there exists $C_*(\|\mathbf{u}_0\|_{H^s}, \nu, T) < \infty$ such that*

$$\int_0^T \|S(t)\|_{L^\infty(\mathbb{T}^3)}\, dt \le C_*(\|\mathbf{u}_0\|_{H^s}, \nu, T), \tag{3.3}$$

where $S = \frac{1}{2}(\nabla\mathbf{u} + (\nabla\mathbf{u})^T)$. Then again $T^* = +\infty$ and $\mathbf{u} \in C^\infty(\mathbb{T}^3 \times (0,\infty))$, by the classical Beale–Kato–Majda continuation criterion [3, 13] (one-directional). The H^s bound is not *the direct exponential (3.2): the Calderón–Zygmund / Riesz relation between $\nabla\mathbf{u}$ and $\boldsymbol{\omega}$ (equivalently S) introduces a logarithmic Sobolev factor [3, 9], and closing the Kato–Ponce estimate produces a double-exponential (or iterated-log) majorant rather than (3.2). We do not claim a direct C_* constant for the strain-only path.*

Remark 3.2 (Comparison of the two forms)**.** Since $\|S\|_{L^\infty} \le \|\nabla\mathbf{u}\|_{L^\infty}$, integrability of $\|\nabla\mathbf{u}\|_{L^\infty}$ implies integrability of $\|S\|_{L^\infty}$: Form (B-grad) implies Form (B-strain). The converse fails at the level of constants: controlling S alone recovers $\nabla\mathbf{u}$ only up to a logarithm, so the H^s majorants differ. Both forms are conditional hypotheses; neither closes Clay Statement (B) unconditionally.

Proof sketch. For (B-grad): the cited Kato–Ponce differential inequality integrates to

$$\|\mathbf{u}(t)\|_{H^s} \le \|\mathbf{u}_0\|_{H^s} + \int_0^t C_s \|\nabla\mathbf{u}(\tau)\|_{L^\infty} \|\mathbf{u}(\tau)\|_{H^s}\, d\tau.$$

Integral Gronwall with (3.1) yields (3.2); the cited $\nabla\mathbf{u}$-continuation criterion then forces $T^* = +\infty$.

For (B-strain): decomposing $\nabla\mathbf{u} = S + \Omega$ gives $(\boldsymbol{\omega}\cdot\nabla)\mathbf{u} = S\boldsymbol{\omega}$. Testing (1.2) against $|\boldsymbol{\omega}|^{p-2}\boldsymbol{\omega}$ and letting $p \to \infty$ produces $\frac{d}{dt}\log M(t) \le \|S(t)\|_{L^\infty}$ for $M = \|\boldsymbol{\omega}\|_{L^\infty}$. Integrating and applying (3.3) bounds M exponentially. The log-Sobolev bound

$$\|\nabla\mathbf{u}(t)\|_{L^\infty} \le C\Big(1 + \|\boldsymbol{\omega}(t)\|_{L^\infty} \log\big(e + \|\mathbf{u}(t)\|_{H^s}\big) + \|\boldsymbol{\omega}(t)\|_{L^2}\Big)$$

[3, 9] feeds into Kato–Ponce and yields only a double-exponential H^s majorant; BKM continuation [3, 13] still gives $T^* = +\infty$. Parabolic bootstrap upgrades to C^∞ for $t > 0$. The converse (smooth on $[0,T] \Rightarrow$ (3.1)/ (3.3)) is the trivial Sobolev embedding $H^s \hookrightarrow W^{1,\infty}$ for $s > 5/2$. □

Theorem 3.3 (Unconditional Local-in-Time Regularity — Kato Lifespan)**.** *For arbitrary large divergence-free initial data $\mathbf{u}_0 \in H^s(\mathbb{T}^3;\mathbb{R}^3)$ ($s > 5/2$), there exists a deterministic lifespan*

$$T_{\mathrm{loc}} \ge \frac{1}{C_s \|\mathbf{u}_0\|_{H^s(\mathbb{T}^3)}}$$

on which Form B holds unconditionally and $\mathbf{u} \in C^\infty(\mathbb{T}^3 \times (0, T_{\mathrm{loc}}))$.

Proof. We record the three standard pillars of the Kato theory [7].
(a) Full Kato–Ponce commutator bound. Applying $\Lambda^s = (I - \Delta)^{s/2}$ to (1.1) and taking the L^2 pairing with $\Lambda^s\mathbf{u}$ produces, after the usual cancellation of the pressure and the divergence-free constraint,

$$\frac{1}{2}\frac{d}{dt}\|\mathbf{u}\|_{H^s}^2 + \nu\|\nabla\mathbf{u}\|_{H^s}^2 \le C_s\Big(\|\nabla\mathbf{u}\|_{L^\infty}\|\mathbf{u}\|_{H^s} + \|\nabla\mathbf{u}\|_{L^\infty}\|\nabla\mathbf{u}\|_{H^{s-1}}\Big)\|\mathbf{u}\|_{H^s}.$$

For $s > 5/2$ the second summand is absorbed into the first by the Sobolev embedding $H^{s-1} \hookrightarrow L^\infty$ on the gradient factor (or by interpolating $\|\nabla\mathbf{u}\|_{H^{s-1}} \le C\|\mathbf{u}\|_{H^s}$), yielding the closed differential inequality

$$\frac{d}{dt}\|\mathbf{u}\|_{H^s} \le C_s\|\mathbf{u}\|_{H^s}^2. \tag{3.4}$$

(b) Existence via mollified/Galerkin approximations. Let $\mathbf{u}_0^\varepsilon = J_\varepsilon \mathbf{u}_0$ be a family of smooth mollifications (or finite-mode Galerkin projections) of the initial datum, with $\|\mathbf{u}_0^\varepsilon\|_{H^s} \le \|\mathbf{u}_0\|_{H^s}$. The corresponding approximate solutions $\mathbf{u}^\varepsilon$ satisfy (3.4) with constants independent of ε. Integrating the Riccati inequality on the common interval

$$\big[0, T_{\mathrm{loc}}\big), \qquad T_{\mathrm{loc}} = \frac{1}{2C_s \|\mathbf{u}_0\|_{H^s}},$$

gives a uniform bound $\sup_{t<T_{\mathrm{loc}}} \|\mathbf{u}^\varepsilon(t)\|_{H^s} \le 2\|\mathbf{u}_0\|_{H^s}$. An Aubin–Lions compactness argument (boundedness in $L^\infty_t H^s$ and of $\partial_t \mathbf{u}^\varepsilon$ in a negative Sobolev space) produces a limit $\mathbf{u} \in L^\infty([0, T_{\mathrm{loc}}); H^s)$ solving (1.1) distributionally; continuity in time with values in H^s follows by the standard energy-identity argument for strong solutions. Parabolic bootstrap then upgrades to $\mathbf{u} \in C^\infty(\mathbb{T}^3 \times (0, T_{\mathrm{loc}}))$.
(c) L^2 uniqueness. If $\mathbf{u}$ and $\mathbf{v}$ are two strong solutions on $[0, T]$ with the same initial datum, set $\mathbf{w} = \mathbf{u} - \mathbf{v}$. Testing the difference of the momentum equations against $\mathbf{w}$ yields

$$\frac{1}{2}\frac{d}{dt}\|\mathbf{w}\|_{L^2}^2 + \nu\|\nabla \mathbf{w}\|_{L^2}^2 = -\int_{\mathbb{T}^3} \big((\mathbf{w}\cdot\nabla)\mathbf{v}\big)\cdot \mathbf{w}\, dV \le \|\nabla \mathbf{v}\|_{L^\infty}\|\mathbf{w}\|_{L^2}^2.$$

Gronwall's lemma with $\mathbf{w}(0) = 0$ forces $\mathbf{w} \equiv 0$. (The gradient factor is that of the second solution $\mathbf{v}$, consistent with the subtraction $\mathbf{u} - \mathbf{v}$.)

Via Sobolev embedding, Form B holds on every compact subinterval of $[0, T_{\mathrm{loc}})$ with

$$C_*(\|\mathbf{u}_0\|_{H^s}, \nu, T) = -\frac{C_{\mathrm{emb}}}{C_s}\log\big(1 - C_s T\|\mathbf{u}_0\|_{H^s}\big) < \infty.$$

□

Theorem 3.4 (Unconditional Global Regularity for Small Critical Data — Fujita–Kato)**.** *There exists an absolute dimensionless constant $\varepsilon > 0$ such that whenever $\mathbf{u}_0 \in H^s(\mathbb{T}^3; \mathbb{R}^3)$ $(s > 5/2)$ satisfies the scale-invariant smallness condition*

$$\|\mathbf{u}_0\|_{\dot H^{1/2}(\mathbb{T}^3)} \le \varepsilon\nu,$$

Form B holds globally on $[0, \infty)$, yielding $T^* = +\infty$ and $\mathbf{u} \in C^\infty(\mathbb{T}^3 \times (0, \infty))$.

Proof. Testing (1.1) against $(-\Delta)^{1/2}\mathbf{u}$ and applying product estimates in homogeneous Sobolev spaces [6] gives

$$\frac{1}{2}\frac{d}{dt}\|\mathbf{u}(t)\|_{\dot H^{1/2}}^2 + \nu\|\mathbf{u}(t)\|_{\dot H^{3/2}}^2 \le C_{\mathrm{FK}}\|\mathbf{u}(t)\|_{\dot H^{1/2}}\|\mathbf{u}(t)\|_{\dot H^{3/2}}^2.$$

Choosing $\varepsilon < 1/(2C_{\mathrm{FK}})$ ensures that $(\nu - C_{\mathrm{FK}}\|\mathbf{u}_0\|_{\dot H^{1/2}}) > \nu/2 > 0$ at $t = 0$, trapping $\|\mathbf{u}(t)\|_{\dot H^{1/2}} \le \|\mathbf{u}_0\|_{\dot H^{1/2}}$ for all $t \ge 0$ and forcing

$$\int_0^\infty \|\mathbf{u}(t)\|_{\dot H^{3/2}}^2\, dt \le \frac{1}{\nu}\|\mathbf{u}_0\|_{\dot H^{1/2}}^2.$$

Interpolating this global $\dot H^{3/2}$ dissipation into higher-order H^s norms yields control of $\|S(t)\|_{L^\infty}$ in $L^1([0, \infty))$ by $\|\mathbf{u}_0\|_{H^s}$ and ν. Theorem 3.1 then yields global smoothness for $t > 0$. □

4. Status Matrix, Lean Scope, and Open Frontier

The companion Lean module formalizes the metriplectic algebra, the degeneracy identity $B(F, H) \equiv 0$, the alignment-kernel / Constantin–Fefferman desingularization bounds, and the integrating-factor Gronwall logic over abstract solution-data interfaces. Analytic Navier–Stokes facts (Clay Statement (B), the BKM criterion, smooth-implies-bounded-strain, and the differential inequality) remain assumed structure fields; the Lean development does not claim an unconditional resolution of the Millennium problem.

TABLE 1. Status matrix for the results recorded in this manuscript.

Item	Status	Content
Theorem A	Conditional	Forms (B-grad)/(B-strain) $\Rightarrow T^* = \infty$ ($t > 0$); (B-grad) gives direct H^s bound; (B-strain) double-exponential via log-Sobolev
Theorem B	Unconditional Local	Kato lifespan; smoothness for $t > 0$
Theorem C	Unconditional Small-Data Global	$\|\mathbf{u}_0\|_{\dot{H}^{1/2}} \leq \varepsilon\nu$
Metriplectic geometry	Exact / Canonical	$\{F, G\}_{\mathrm{LP}}$ skew; $[F, G]_{\mathrm{diss}} + B$ symmetric; $B(F, H) \equiv 0$
BKM continuation	One-directional	$\int_0^T \|\nabla\mathbf{u}\|_{L^\infty}\, dt < \infty \implies T^* > T$ (also via $\|S\|_{L^\infty}$, with log loss)
Lean formalization	Machine-checked (conditional)	Metriplectic algebra, degeneracy, alignment desingularization, Gronwall logic over abstract PDE input types; 0 `sorry`; axioms `[propext, Classical.choice, Quot.sound]`; Lean / Mathlib `v4.30.0-rc1`
Clay Statement (B)	Open	The $L^2_t L^2_x \to L^1_t L^\infty_x$ scaling gap

4.1. **Isolation of the $L^2_t L^2_x \to L^1_t L^\infty_x$ Scaling Gap.** As demonstrated by Theorems 3.1–3.4, the large-data global regularity problem reduces to establishing a data-only *a priori* bound on $\int_0^T \|\nabla\mathbf{u}(t)\|_{L^\infty}\, dt$ or $\int_0^T \|S(t)\|_{L^\infty}\, dt$ (Forms B-grad / B-strain). The phase-space geometry makes transparent why standard energy methods cannot bridge this gap:

(1) **Supercritical Dissipation vs. Critical Strain.** The only coercive global bound generated unconditionally by $[F, H]_{\mathrm{diss}}$ is the $L^2_t L^2_x$ enstrophy dissipation (1.3), which scales as λ^{-1} under parabolic rescaling. Forms B-grad / B-strain require $L^1_t L^\infty_x$ control (λ^0).

(2) **Necessity of Viscous Coupling.** Because $B(F, H) \equiv 0$ leaves the reversible Euler sector untouched—and inviscid 3D Euler admits finite-time singularities in low-regularity classes—no purely symplectic orbit invariant can close Form B-grad / B-strain without exploiting viscous dissipation.

4.2. **Directions for Geometric-Coherence Analysis.** While the current degenerate bracket $B(F, G)$ preserves the unmodified flow without injecting artificial damping, its projected structure $\Pi_{\mathbf{u}}$ points toward geometric regularity criteria. Expressing the strain via the Constantin–Fefferman singular integral kernel [4] shows that vortex stretching cancels when the vorticity direction $\boldsymbol{\xi} = \boldsymbol{\omega}/|\boldsymbol{\omega}|$ remains Lipschitz-coherent in intense regions. Whether localized coherence bounds on $\boldsymbol{\xi}$ can be coupled to $[F, G]_{\mathrm{diss}}$ to bridge the scaling gap for arbitrary large data remains open.

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